10TH CLASS MATHEMATICS — NEW SYLLABUS (PECTAA)

Exercise 3.4 — Multiplication of Matrices

Unit 3 | Solved Exercise | New Book (PECTAA)

Exercise 3.4 (Solved)

1. Find AB and BA (if possible) for the following pairs of matrices:

Two matrices can be multiplied only if the number of columns of the first matrix is equal to the number of rows of the second. To get the element in row i and column j of AB, multiply row i of A with column j of B and add.

(i) $A = \begin{bmatrix} 1 & 2 \\ -1 & 0 \end{bmatrix}$ **and** $B = \begin{bmatrix} 3 & 2 \\ 1 & -1 \end{bmatrix}$

$$AB = \begin{bmatrix} 1 & 2 \\ -1 & 0 \end{bmatrix} \times \begin{bmatrix} 3 & 2 \\ 1 & -1 \end{bmatrix}$$

A is 2 × 2 and B is 2 × 2, so AB is also 2 × 2

Row 1 × Column 1 = (1)(3) + (2)(1) = 5

Row 1 × Column 2 = (1)(2) + (2)(−1) = 0

Row 2 × Column 1 = (−1)(3) + (0)(1) = −3

Row 2 × Column 2 = (−1)(2) + (0)(−1) = −2

$$AB = \begin{bmatrix} 5 & 0 \\ -3 & -2 \end{bmatrix}$$

$$BA = \begin{bmatrix} 3 & 2 \\ 1 & -1 \end{bmatrix} \times \begin{bmatrix} 1 & 2 \\ -1 & 0 \end{bmatrix}$$

Row 1 × Column 1 = (3)(1) + (2)(−1) = 1

Row 1 × Column 2 = (3)(2) + (2)(0) = 6

Row 2 × Column 1 = (1)(1) + (−1)(−1) = 2

Row 2 × Column 2 = (1)(2) + (−1)(0) = 2

$$BA = \begin{bmatrix} 1 & 6 \\ 2 & 2 \end{bmatrix}$$

Here AB ≠ BA, so matrix multiplication is not commutative

Answer: $AB = \begin{bmatrix} 5 & 0 \\ -3 & -2 \end{bmatrix}$, $BA = \begin{bmatrix} 1 & 6 \\ 2 & 2 \end{bmatrix}$

(ii) $A = \begin{bmatrix} 1 & -2 \end{bmatrix}$ **and** $B = \begin{bmatrix} 3 \\ -4 \end{bmatrix}$

studysapiens.com

A is 1 × 2 and B is 2 × 1, so AB is of order 1 × 1

$$AB = \begin{bmatrix} 1 & -2 \end{bmatrix} \times \begin{bmatrix} 3 \\ -4 \end{bmatrix}$$

Row 1 × Column 1 = (1)(3) + (−2)(−4) = 3 + 8 = 11

$$AB = \begin{bmatrix} 11 \end{bmatrix}$$

B is 2 × 1 and A is 1 × 2, so BA is of order 2 × 2

$$BA = \begin{bmatrix} 3 \\ -4 \end{bmatrix} \times \begin{bmatrix} 1 & -2 \end{bmatrix} = \begin{bmatrix} 3 & -6 \\ -4 & 8 \end{bmatrix}$$

Answer: $AB = \begin{bmatrix} 11 \end{bmatrix}$, $BA = \begin{bmatrix} 3 & -6 \\ -4 & 8 \end{bmatrix}$

(iii) $A = \begin{bmatrix} 4 \\ 4 \end{bmatrix}$ **and** $B = \begin{bmatrix} 2 & 5 \end{bmatrix}$

A is 2 × 1 and B is 1 × 2, so AB is of order 2 × 2

$$AB = \begin{bmatrix} 4 \\ 4 \end{bmatrix} \times \begin{bmatrix} 2 & 5 \end{bmatrix} = \begin{bmatrix} 8 & 20 \\ 8 & 20 \end{bmatrix}$$

Row 1 × Column 1 = (4)(2) = 8, Row 1 × Column 2 = (4)(5) = 20

Same way row 2 also gives 8 and 20

B is 1 × 2 and A is 2 × 1, so BA is of order 1 × 1

$$BA = \begin{bmatrix} 2 & 5 \end{bmatrix} \times \begin{bmatrix} 4 \\ 4 \end{bmatrix}$$

Row 1 × Column 1 = (2)(4) + (5)(4) = 8 + 20 = 28

$$BA = \begin{bmatrix} 28 \end{bmatrix}$$

Answer: $AB = \begin{bmatrix} 8 & 20 \\ 8 & 20 \end{bmatrix}$, $BA = \begin{bmatrix} 28 \end{bmatrix}$

(iv) $A = \begin{bmatrix} 1 & 2 \\ -1 & 1 \\ 3 & 0 \end{bmatrix}$ **and** $B = \begin{bmatrix} -1 & 4 & 1 \\ 2 & 3 & 1 \end{bmatrix}$

A is 3 × 2 and B is 2 × 3, so AB is of order 3 × 3

studysapiens.com

$$AB = \begin{bmatrix} 1 & 2 \\ -1 & 1 \\ 3 & 0 \end{bmatrix} \times \begin{bmatrix} -1 & 4 & 1 \\ 2 & 3 & 1 \end{bmatrix}$$

Row 1 × Column 1 = (1)(−1) + (2)(2) = −1 + 4 = 3

Row 1 × Column 2 = (1)(4) + (2)(3) = 4 + 6 = 10

Row 1 × Column 3 = (1)(1) + (2)(1) = 1 + 2 = 3

Row 2 × Column 1 = (−1)(−1) + (1)(2) = 1 + 2 = 3

Row 2 × Column 2 = (−1)(4) + (1)(3) = −4 + 3 = −1

Row 2 × Column 3 = (−1)(1) + (1)(1) = −1 + 1 = 0

Row 3 × Column 1 = (3)(−1) + (0)(2) = −3

Row 3 × Column 2 = (3)(4) + (0)(3) = 12

Row 3 × Column 3 = (3)(1) + (0)(1) = 3

$$AB = \begin{bmatrix} 3 & 10 & 3 \\ 3 & -1 & 0 \\ -3 & 12 & 3 \end{bmatrix}$$

Now B is 2 × 3 and A is 3 × 2, so BA is of order 2 × 2

$$BA = \begin{bmatrix} -1 & 4 & 1 \\ 2 & 3 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 2 \\ -1 & 1 \\ 3 & 0 \end{bmatrix}$$

Row 1 × Column 1 = (−1)(1) + (4)(−1) + (1)(3) = −1 − 4 + 3 = −2

Row 1 × Column 2 = (−1)(2) + (4)(1) + (1)(0) = −2 + 4 = 2

Row 2 × Column 1 = (2)(1) + (3)(−1) + (1)(3) = 2 − 3 + 3 = 2

Row 2 × Column 2 = (2)(2) + (3)(1) + (1)(0) = 4 + 3 = 7

$$BA = \begin{bmatrix} -2 & 2 \\ 2 & 7 \end{bmatrix}$$

Answer: $AB = \begin{bmatrix} 3 & 10 & 3 \\ 3 & -1 & 0 \\ -3 & 12 & 3 \end{bmatrix}$, $BA = \begin{bmatrix} -2 & 2 \\ 2 & 7 \end{bmatrix}$

(v) $A = \begin{bmatrix} 1 & 2 & 2 \\ 3 & 1 & 1 \end{bmatrix}$ **and** $B = \begin{bmatrix} 1 & 5 \\ 2 & 4 \\ 1 & 6 \end{bmatrix}$

A is 2 × 3 and B is 3 × 2, so AB is of order 2 × 2

$$AB = \begin{bmatrix} 1 & 2 & 2 \\ 3 & 1 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 5 \\ 2 & 4 \\ 1 & 6 \end{bmatrix}$$

Row 1 × Column 1 = (1)(1) + (2)(2) + (2)(1) = 1 + 4 + 2 = 7

Row 1 × Column 2 = (1)(5) + (2)(4) + (2)(6) = 5 + 8 + 12 = 25

Row 2 × Column 1 = (3)(1) + (1)(2) + (1)(1) = 3 + 2 + 1 = 6

Row 2 × Column 2 = (3)(5) + (1)(4) + (1)(6) = 15 + 4 + 6 = 25

$$AB = \begin{bmatrix} 7 & 25 \\ 6 & 25 \end{bmatrix}$$

B is 3 × 2 and A is 2 × 3, so BA is of order 3 × 3

$$BA = \begin{bmatrix} 1 & 5 \\ 2 & 4 \\ 1 & 6 \end{bmatrix} \times \begin{bmatrix} 1 & 2 & 2 \\ 3 & 1 & 1 \end{bmatrix}$$

Row 1 × Column 1 = (1)(1) + (5)(3) = 16, Row 1 × Column 2 = (1)(2) + (5)(1) = 7

Row 2 × Column 1 = (2)(1) + (4)(3) = 14, Row 2 × Column 2 = (2)(2) + (4)(1) = 8

Row 3 × Column 1 = (1)(1) + (6)(3) = 19, Row 3 × Column 2 = (1)(2) + (6)(1) = 8

$$BA = \begin{bmatrix} 16 & 7 & 7 \\ 14 & 8 & 8 \\ 19 & 8 & 8 \end{bmatrix}$$

Answer: $AB = \begin{bmatrix} 7 & 25 \\ 6 & 25 \end{bmatrix}$, $BA = \begin{bmatrix} 16 & 7 & 7 \\ 14 & 8 & 8 \\ 19 & 8 & 8 \end{bmatrix}$

2. If A , B and C are the following matrices, then verify:

$$A = \begin{bmatrix} 5 & 1 \\ -1 & 4 \end{bmatrix} \quad B = \begin{bmatrix} 2 & -1 \\ 0 & 3 \end{bmatrix} \quad C = \begin{bmatrix} 5 & 1 \\ 1 & 4 \end{bmatrix}$$

(i) $AB \neq BA$

(ii) $A(B - C) = AB - AC$

(iii) $A(BC) = (AB)C$

(iv) $(BC)^t = C^tB^t$

(v) $(B + C)A = BA + CA$

Products needed

$$AB = \begin{bmatrix} 5 & 1 \\ -1 & 4 \end{bmatrix} \times \begin{bmatrix} 2 & -1 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} 10 & -2 \\ -2 & 13 \end{bmatrix}$$

$$BA = \begin{bmatrix} 2 & -1 \\ 0 & 3 \end{bmatrix} \times \begin{bmatrix} 5 & 1 \\ -1 & 4 \end{bmatrix} = \begin{bmatrix} 11 & -2 \\ -3 & 12 \end{bmatrix}$$

$$AC = \begin{bmatrix} 5 & 1 \\ -1 & 4 \end{bmatrix} \times \begin{bmatrix} 5 & 1 \\ 1 & 4 \end{bmatrix} = \begin{bmatrix} 26 & 9 \\ -1 & 15 \end{bmatrix}$$

$$BC = \begin{bmatrix} 2 & -1 \\ 0 & 3 \end{bmatrix} \times \begin{bmatrix} 5 & 1 \\ 1 & 4 \end{bmatrix} = \begin{bmatrix} 9 & -2 \\ 3 & 12 \end{bmatrix}$$

$$B + C = \begin{bmatrix} 7 & 0 \\ 1 & 7 \end{bmatrix} \qquad B - C = \begin{bmatrix} -3 & -2 \\ -1 & -1 \end{bmatrix}$$

(i) $AB \neq BA$

$$AB = \begin{bmatrix} 10 & -2 \\ -2 & 13 \end{bmatrix}$$

$$BA = \begin{bmatrix} 11 & -2 \\ -3 & 12 \end{bmatrix}$$

The corresponding elements are not equal, so $AB \neq BA$

Answer: $AB \neq BA$ (verified)

(ii) $A(B - C) = AB - AC$

$$\text{L.H.S.} = A(B - C) = \begin{bmatrix} 5 & 1 \\ -1 & 4 \end{bmatrix} \times \begin{bmatrix} -3 & -2 \\ -1 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} -16 & -11 \\ -1 & -2 \end{bmatrix}$$

$$\text{R.H.S.} = AB - AC = \begin{bmatrix} 10 & -2 \\ -2 & 13 \end{bmatrix} - \begin{bmatrix} 26 & 9 \\ -1 & 15 \end{bmatrix}$$

$$= \begin{bmatrix} -16 & -11 \\ -1 & -2 \end{bmatrix}$$

Answer: L.H.S. = R.H.S., so $A(B - C) = AB - AC$ (verified)

studysapiens.com

(iii) A(BC) = (AB)C

$$\text{L.H.S.} = A(BC) = \begin{bmatrix} 5 & 1 \\ -1 & 4 \end{bmatrix} \times \begin{bmatrix} 9 & -2 \\ 3 & 12 \end{bmatrix}$$

$$= \begin{bmatrix} 48 & 2 \\ 3 & 50 \end{bmatrix}$$

$$\text{R.H.S.} = (AB)C = \begin{bmatrix} 10 & -2 \\ -2 & 13 \end{bmatrix} \times \begin{bmatrix} 5 & 1 \\ 1 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} 48 & 2 \\ 3 & 50 \end{bmatrix}$$

Answer: L.H.S. = R.H.S., so A(BC) = (AB)C (verified)

(iv) $(BC)^t = C^tB^t$

$$\text{L.H.S.} = (BC)^t = \begin{bmatrix} 9 & 3 \\ -2 & 12 \end{bmatrix}$$

$$C^t = \begin{bmatrix} 5 & 1 \\ 1 & 4 \end{bmatrix} \quad B^t = \begin{bmatrix} 2 & 0 \\ -1 & 3 \end{bmatrix}$$

$$\text{R.H.S.} = C^tB^t = \begin{bmatrix} 5 & 1 \\ 1 & 4 \end{bmatrix} \times \begin{bmatrix} 2 & 0 \\ -1 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 9 & 3 \\ -2 & 12 \end{bmatrix}$$

Answer: L.H.S. = R.H.S., so $(BC)^t = C^tB^t$ (verified)

studysapiens.com

(v) (B + C)A = BA + CA

$$\text{L.H.S.} = (B + C)A = \begin{bmatrix} 7 & 0 \\ 1 & 7 \end{bmatrix} \times \begin{bmatrix} 5 & 1 \\ -1 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} 35 & 7 \\ -2 & 29 \end{bmatrix}$$

$$CA = \begin{bmatrix} 5 & 1 \\ 1 & 4 \end{bmatrix} \times \begin{bmatrix} 5 & 1 \\ -1 & 4 \end{bmatrix} = \begin{bmatrix} 24 & 9 \\ 1 & 17 \end{bmatrix}$$

$$\text{R.H.S.} = BA + CA = \begin{bmatrix} 11 & -2 \\ -3 & 12 \end{bmatrix} + \begin{bmatrix} 24 & 9 \\ 1 & 17 \end{bmatrix}$$

$$= \begin{bmatrix} 35 & 7 \\ -2 & 29 \end{bmatrix}$$

Answer: L.H.S. = R.H.S., so (B + C)A = BA + CA (verified)

3. If [[4, a], [b, 3]] × [[6], [6]] = [[6], [3]], then find the values of a and b.

Solution

$$\begin{bmatrix} 4 & a \\ b & 3 \end{bmatrix} \times \begin{bmatrix} 6 \\ 6 \end{bmatrix} = \begin{bmatrix} 6 \\ 3 \end{bmatrix}$$

Multiplying the left side, row 1 × column 1 gives

$(4)(6) + (a)(6) = 6$

$24 + 6a = 6$

$6a = 6 - 24$

$= -18$

$a = -18 \div 6$

$= -3$

Row 2 × column 1 gives

$(b)(6) + (3)(6) = 3$

$6b + 18 = 3$

$6b = 3 - 18$

$= -15$

$b = -15 \div 6$

$= -5/2$

Answer: a = −3, b = −5/2

studysapiens.com

4. If [[x, 1], [y, 2]] × [[1, 0], [3, −1]] = [[7, −1], [4, −2]], then find the values of x and y.

Solution

$$\begin{bmatrix} x & 1 \\ y & 2 \end{bmatrix} \times \begin{bmatrix} 1 & 0 \\ 3 & -1 \end{bmatrix} = \begin{bmatrix} 7 & -1 \\ 4 & -2 \end{bmatrix}$$

Row 1 of the product: (x)(1) + (1)(3) = 7

x + 3 = 7

x = 7 − 3

= 4

Check the second element of row 1: (x)(0) + (1)(−1) = −1, which is correct

Row 2 of the product: (y)(1) + (2)(3) = 4

y + 6 = 4

y = 4 − 6

= −2

Check the second element of row 2: (y)(0) + (2)(−1) = −2, which is correct

Answer: x = 4, y = −2

studysapiens.com